

RESEARCH ARTICLE

Neural Sensitivity to Social Media Use: Implications for Sleep Duration in Adolescents

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ABSTRACT

Sleep plays a foundational role in adolescent development, supporting emotional regulation, cognition and brain maturation. However, adolescents today face increasing challenges to healthy sleep partly due to widespread social media use and heightened reward sensitivity. While each factor has been studied independently, less is known about how they interact, especially whether brain activation shapes how social media influences sleep. We examined prospective associations between self-reported sleep duration, social media use and brain activation in 1985 adolescents (mean age, Year 2 = 11.91 years) from the Adolescent Brain Cognitive Development (ABCD) Study. Sleep duration was measured via the Munich Chronotype Questionnaire; social media use via the Youth Screen Time Survey and brain activity using fMRI during the Monetary Incentive Delay task. We tested three sets of prospective models to examine how sleep, social media use, executive control and reward-related brain activation influence each other over two years, controlling sociodemographics. Greater social media use at Year 2 predicted shorter sleep duration at Year 4, particularly among adolescents with lower activation in the nucleus accumbens, cingulate gyrus, insula and putamen. Shorter sleep also predicted increased social media use two years later, with effects moderated by activation in the insula and middle frontal gyrus. Longer sleep at Year 2 predicted higher caudate activation at Year 4. Brain activation in reward and executive control-related regions moderated associations between sleep and social media use, suggesting that lower neural engagement may reflect increased susceptibility to the sleep-disrupting effects of social media.

1 | Introduction

Adolescence is a dynamic period of development characterized by profound changes in brain maturation, social behavior, emotional regulation and sleep (Dahl 2004). Biologically, adolescent sleep changes are marked by shifts in circadian rhythms, contributing to a preference for later bedtimes and wake up times (Galván 2020), which leads to a mismatch with early school start times and increasing academic and social pressures. Consequently, adolescents often obtain insufficient sleep (Short

et al. 2013). Sleep facilitates restorative processes, memory consolidation and neural plasticity (Telzer et al. 2013), while also playing a key role in emotional regulation and social functioning, and insufficient sleep can therefore negatively impact a wide range of functions in adolescents.

Neurocognitively, the brain matures structurally and functionally, influencing executive control and reward processing (Casey et al. 2005). Brain regions involved in reward processing and executive control mature at different rates during adolescence,

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a concept central to the dual systems model (Holm et al. 2009). During adolescence, there is a natural increase in reward sensitivity and a slower rate of cognitive development (Steinberg 2010). Regions important for reward processing and executive control include, but are not limited to, the nucleus accumbens, amygdala, cingulate cortex, insula, caudate, precuneus, putamen, thalamus and inferior, middle and superior frontal gyri (Perogamvros and Schwartz 2012). Changes in the reward system during adolescence contribute to improved cognitive abilities (Murty et al. 2016). These developments are shaped by underlying biological mechanisms such as synaptic pruning, which refines neural circuits based on experience and use (Larsen and Luna 2018). While the pace of development varies across individuals and brain regions, many of these neurodevelopmental processes continue into the early to mid-20s (Yurgelun-Todd 2007).

Socially, adolescents gain increasing independence and expand their social interactions both in person and digitally (Sanders 2013). In today's digital landscape, social media is deeply embedded in the lives of children and adolescents. Although most platforms require users to be at least 13 years old, 38% of children aged 8–12 already have at least one social media account (Rideout and Robb 2019). Younger adolescents (ages 8–12) spend an average of 4 h and 44 min per day on screen-related activities, while teens aged 13–18 average 7 h and 22 min daily (Rideout and Robb 2019). Despite potential benefits, social media use has been consistently associated with various negative outcomes in adolescents, including sleep problems and mood disorders such as anxiety and depression (Keles et al. 2020).

Limited research has examined the interconnected relationships between sleep, brain activation and social media use in adolescents. For example, adolescents with poor sleep exhibit heightened activation in the insula and ventral striatum when responding to monetary rewards (Holm et al. 2009; Mullin et al. 2013). Poor sleep quality, shorter sleep duration and later sleep timing have also been associated with decreased activation in other reward-related regions, such as the caudate and medial prefrontal cortex (Holm et al. 2009; Hasler et al. 2012). Although the reverse direction is less studied, Perogamvros and Schwartz (2012) propose that reward-related neural activity during sleep may influence sleep architecture, particularly REM sleep, through emotional processing. It remains unclear whether there is a reciprocal interaction between reward processing and sleep processes during adolescence.

Furthermore, a bidirectional relationship between sleep and social media use has been proposed (Hale and Guan 2015), suggesting that increased screen time including social media, is linked to shorter sleep duration, while insufficient sleep may also lead to greater social media engagement. Given that mobile devices are often used during the evening and nighttime hours, social media use around bedtime or during the night can displace sleep (Cain and Gradisar 2010). Additionally, the mental and physical stimulation from the content and social interactions may make it harder to fall asleep and stay asleep (Hale and Guan 2015).

There is evidence for a bidirectional relationship between executive control, reward processing and social media use

during adolescence. Brain regions, including those related to self-regulation, decision making and social interactions, which develop across adolescence, influence choices about the timing, frequency and intensity of social media engagement (Crone and Konijn 2018). Concurrently, social media offers stimulating content that provides instant gratification through dopamine release (Mujica et al. 2022). Adolescents who receive more 'likes' on their social media posts exhibit greater activation in brain regions involved in reward processing (Sherman et al. 2016). Online social network size has also been associated with structural changes in the brain, including increased gray matter volume in the right superior temporal sulcus, in the left middle temporal gyrus and in the entorhinal cortex (Kanai et al. 2012). The continuous availability of rewarding online interactions may reinforce habitual use by activating reward pathways (Wadsley et al. 2022), creating a feedback loop that could challenge adolescents' developing executive control.

In summary, social media use, executive control- and reward-related brain activation and sleep patterns appear to be interconnected, though the direction and specific mechanisms of these associations are not fully understood. To date, no study has simultaneously explored these three factors or examined their potential interactive effects. Here, we investigated the directional associations between sleep, social media use and neural responses to reward over time in adolescents. We hypothesized that greater social media use would be associated with shorter sleep duration, moderated by individual differences in brain activation within executive control and reward-processing regions, suggesting that adolescents may differ in their neural sensitivity to social media engagement in ways that impact their sleep. We also examined whether sleep duration and neural activation predicted later social media use. This allowed us to test whether insufficient sleep is associated with greater digital engagement and how social media use and sleep duration are related to executive control and reward-related brain responses, both independently and interactively.

2 | Methods

2.1 | Sample

Data were collected from the Adolescent Brain Cognitive Development (ABCD) Study (Data Release 5.1), an ongoing, multi-site longitudinal research project that enrolled participants, aged 9–10 years at baseline, from 21 different sites across the United States. Both primary caregivers and youth provided informed consent and assent. The study protocol was approved by a centralized institutional review board (University of California, San Diego). Full details about the study can be found in Casey et al. (2018). In the 5.1 release, only half of the Year 4 data was available. Out of the 2270 participants who had neuroimaging data, self-reported sleep and social media data from both Years 2 and 4, 265 participants were excluded for image quality issues, and 20 participants were excluded due to missing covariates. A total of 1985 adolescents with Years 2 and 4 data (age range 10.75–13.75; Mean age, Year 2 = 11.91 years, SD = 0.65) were included in the analyses (see Figure 1 and Table 1). To mitigate potential

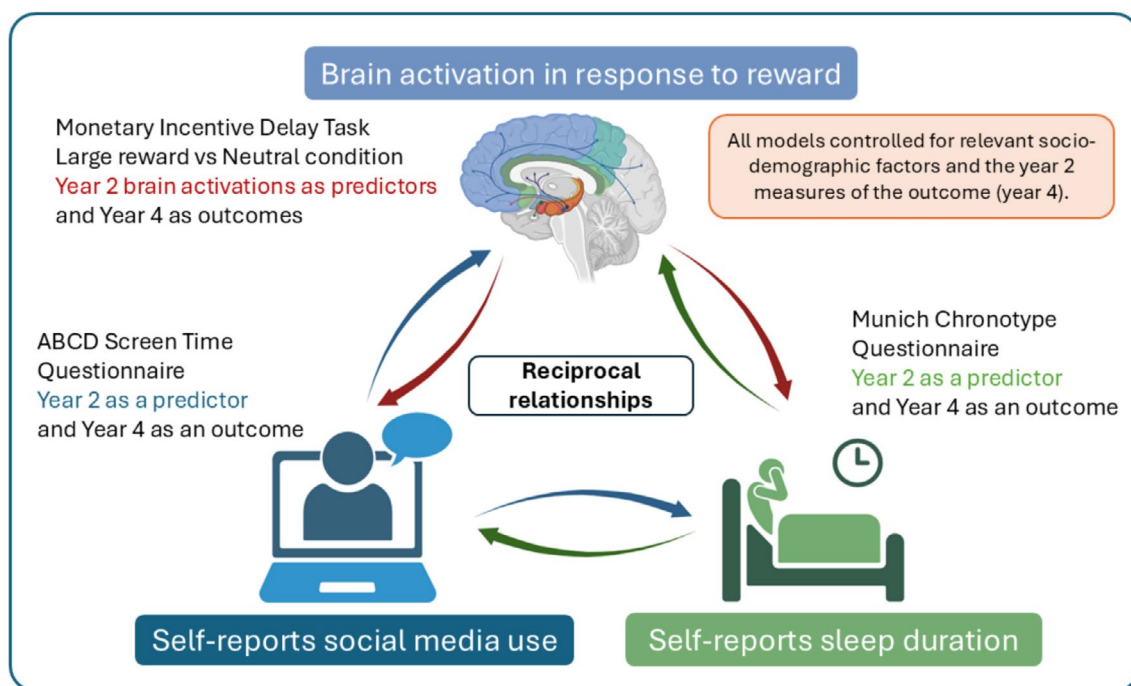


FIGURE 1 | Timeline of data collection in the ABCD study and graphical representation of the statistical model. Sociodemographic data were collected at baseline. MRI data were collected in Years 2 and 4. The Munich Chronotype Questionnaire (MCTQ) was administered in Years 2, 3 and 4 to assess sleep patterns, and the self-reported social media data were aligned with these timepoints. We used predictors from Year 2 to predict outcomes from Year 4.

TABLE 1 | Demographics collected at baseline for the subset of ABCD Study participants included in the current analysis.

Variable	Analytical sample, N=1985
Age at Year 2 (years), mean (range)	11.91 (10–14)
Sex	
Female	900 (45.4%)
Male	1085 (54.6%)
Race/ethnicity	
Asian	38 (1.9%)
Black	185 (9.3%)
Latino/Hispanic	408 (20.5%)
Other	204 (10.2%)
White	1150 (57.9%)
Parental education	
< High school diploma or less	216 (10.9%)
College or above	1769 (89.1%)

selection bias in the ABCD sampling and recruitment process, we employed propensity weights, calibrating ABCD distributions to nationally representative controls from the American Community Survey.

2.2 | Measures

Sleep Duration. Adolescents reported their sleep duration using the Munich Chronotype Questionnaire (Roenneberg et al. 2003), with data from Years 2 and 4 included in the analysis. Sleep duration was derived from self-reported sleep onset and offset times, without accounting for wakefulness after sleep onset. To account for differences between school days and weekends, a weighted sleep duration was calculated ($[\text{weekday sleep duration} \times 5] + [\text{weekend sleep duration} \times 2]$)/7. Values were winsorized at the 1st and 99th percentiles to minimize the influence of extreme outliers.

Social Media Use. Adolescents self-reported their social media usage at Years 2 and 4 using the ABCD Screen Time Questionnaire (Bagot et al. 2022). They provided estimates of their typical daily hours spent on social media, separately for school days and weekend days, excluding school-related activities. To obtain an overall measure of social media use, we calculated a weighted average based on the number of weekdays (5) and weekend days (2) per week ($[\text{weekday social media usage} \times 5] + [\text{weekend social media usage} \times 2]$)/7. Values were winsorized at the 99th percentiles to minimize the influence of extreme outliers. Some participants reported no social media use, which is plausible given their age; therefore, we kept the original values at the lower end of the distribution.

Functional MRI Measures. As part of the ABCD study protocol, adolescents completed multi-modal MRI scans every second year, including Years 2 and 4. Neuroimaging data were collected across multiple study sites using 3T MRI scanners from three different manufacturers: Siemens, GE and Philips.

The monetary incentive delay (MID) task assesses reward processing by measuring neural responses to anticipated monetary gains and losses (Knutson et al. 2000). Participants were trained on the task and completed a practice run before two experimental runs in the scanner. Trial conditions were cued at the start of each trial: circles indicated a potential monetary gain (small: \$0.20, large: \$5.00), squares indicated a potential monetary loss (small: -\$0.20, large: -\$5.00) and triangles represented neutral trials with no financial consequence. Participants responded as quickly as possible to a target presented after a variable anticipation period. Each session includes 50 trials across 5 min and 42 s. Participants' pay reflects task performance, averaging \$20.

Preprocessed fMRI data were obtained from the ABCD 5.1 release, with all imaging data centrally processed by Data Analysis, Informatics and Resource Center to ensure methodological consistency and quality control. The process includes running 1st level analyses on the BOLD time series to estimate activation beta-weights. We analyzed beta-weights in subcortical and cortical regions of interest (ROIs) for each hemisphere using Destrieux parcellation, after merging Run1 and Run2 beta-weights using a weighted-average, with weights determined by the number of frames surviving quality control for each run. Our primary contrasts examined the activation differences between large reward vs. neutral conditions. Executive control and reward-related areas ROIs included the nucleus accumbens, amygdala, cingulate cortex, insula, caudate, precuneus, putamen, thalamus proper, hippocampus and inferior, middle and superior frontal gyri. Mean bilateral activation values were computed for each ROI.

Covariates. Youth sex (female/male), race/ethnicity (White, Black, Hispanic, Asian and Other), age (Year 2), parental education (some college or above, high school or less) and the date of data collection at Year 2 were included in all analyses. To control for the potential impact of the COVID-19 pandemic, the date of data collection at Year 2 was coded as 0 = before February 2020, and 1 = after February 2020, based on WHO's declaration of COVID-19 as a global health emergency.

2.3 | Analytical Plan

To assess the coherence of brain activation across selected regions, we conducted a Confirmatory Factor Analysis (CFA) using data from Year 2. Activation in executive control areas (inferior frontal gyrus, middle frontal gyrus, superior frontal gyrus and precuneus) and reward-related areas (accumbens, amygdala, caudate, hippocampus, putamen, cingulate gyrus and insular gyrus) were included as separate factors. Model fit was evaluated using the Comparative Fit Index (CFI), Tucker-Lewis Index (TLI), Root Mean Square Error of Approximation (RMSEA) and Standardized Root Mean Square Residual (SRMR).

We conducted three sets of prospective models to examine the relationships between sleep duration, social media use and brain activation in executive control and reward-processing ROIs. Within each set, brain activation predictors were included separately, resulting in a total of 12 models per set. In the first

set, we tested the direct and interaction effects of social media usage and brain activation on sleep duration. In the second set, we tested the direct and interaction effects of time spent on social media and sleep duration on brain activation. In the third set, we tested the direct and interaction effects of sleep duration and brain activation on time spent on social media.

All models controlled for relevant covariates, including sociodemographic measures and pandemic status at Year 2. All models were prospective, with Year 2 predictors used to predict Year 4 outcomes, while controlling for the respective Year 2 measure of the outcome variable. Additionally, all models included family ID (to account for clustering effects of participants within families) and MRI scanner manufacturer ID as random terms.

After fitting the models, we calculated the average R^2 across models within each set. Based on these values, we prioritized the interpretation of the models with the highest explained variance. To account for multiple comparisons, we applied False Discovery Rate correction on each set of models, and we reported corrected p -values (at $p < 0.05$) in the results. The analyses were carried out in R and Python 3.5. For the first set of models, we applied a natural log ($x + 1$) transformation to time spent on social media use (both Year 4, the outcome and Year 2, which was included as a covariate) to address skewness (original skewness Year 4: 2.55; Year 2: 5.67). Given the distribution of social media use, we fitted a glmmTMB (Brooks et al. 2017) model to separately account for participants who reported no social media use (zero-inflated component) and those who had social media exposure. In all other models, we used lmer (Bates et al. 2015) to estimate linear mixed-effects models. Furthermore, all continuous predictors were standardized (mean = 0, SD = 1) prior to inclusion in the models.

2.4 | Cross-Validated Model Performance and Statistical Comparison

We used 10-fold cross-validation to estimate out-of-sample predictive accuracy for each model and computed cross-validated R^2 values for each ROI. To ensure that the models converge and the model fit remains comparable, we did not include the random term structure, but all other covariates from the main analyses were integrated. We compared predictive performance across three analytic approaches. These included models predicting sleep duration in Year 4, models predicting social media use in Year 4, and models predicting executive controlling and reward-related areas in Year 4. We used a Friedman rank-sum test to evaluate differences across these approaches. This non-parametric approach avoids distributional assumptions about R^2 values and accounts for the repeated-measures structure. When a significant overall effect was observed, we conducted post hoc Wilcoxon signed-rank tests with Holm correction for multiple comparisons. Statistical significance was set at $p < 0.01$.

3 | Results

In Year 4, when participants were approximately 14 years old, 4% ($n = 79$) reported sleeping less than 6 h of sleep, 26.8% ($n = 611$)

reported sleeping between 6 and 8 h, while 57.7% ($n=1145$) slept between 8 and 9.99 h, and 11.5% ($n=229$) slept 10 h or more hours. In Year 4, 33.6% ($n=666$) of participants reported spending more than 120 min per day on social media, while 26.3% ($n=523$) reported no social media use. Additionally, 20.4% ($n=405$) reported spending between 60 and 119 min, and 19.7% ($n=391$) reported spending between 1 and 59.9 min per day on social media in Year 4.

Testing the coherence of activations across ROIs, the results of the confirmatory factor analysis suggest that executive control areas (inferior frontal gyrus, middle frontal gyrus, superior frontal gyrus, and precuneus) and reward-related areas (accumbens, amygdala, caudate, hippocampus, putamen, cingulate gyrus, and insular gyrus) were clustered together (see [Supporting Information](#) for more details).

3.1 | Executive Control and Reward-Related Brain Activation and Social Media Usage Predicting Sleep Duration

Social media use from Year 2 was significantly associated with shorter sleep duration in Year 4 ($\chi^2=7.76$, $p=0.016$) (Figure 2). Although brain activity (in Year 2) did not show a main effect, significant interaction effects were observed between social media use (Year 2) and activation in Year 2 within the nucleus accumbens ($\chi^2=7.11$, $p=0.021$), cingulate gyrus ($\chi^2=6.61$, $p=0.027$), insula ($\chi^2=9.15$, $p=0.011$) and putamen ($\chi^2=7.13$, $p=0.021$) on sleep duration at Year 4. Higher social media (Year 2) use was associated with shorter sleep duration (Year 4), particularly among participants with lower activation in these brain regions (Figure 3). Year 2 sleep duration was positively related to Year 4 sleep duration ($\chi^2=189.82$, $p<0.01$). Age was also a predictor in the model, with older adolescents reporting shorter sleep ($\chi^2=12.03$, $p=0.002$).

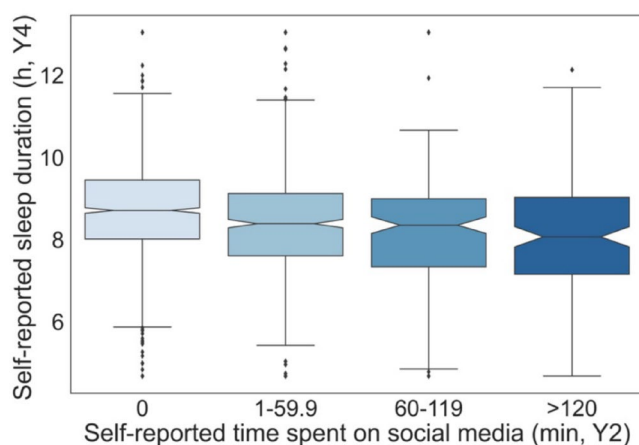


FIGURE 2 | Association between self-reported sleep duration (Y2) and self-reported time spent on social media (Y4). The notches in the boxes represent confidence intervals around the median, and the whiskers indicate the data range excluding outliers. Self-reported social media use was grouped into categories for visualization purposes. Social media use from Year 2 was significantly associated with shorter sleep duration in Year 4 ($\chi^2=7.76$, $p=0.016$).

3.2 | Sleep Duration, Executive Control and Reward-Related Brain Activation Predicting Social Media Use

Shorter sleep duration in Year 2 was significantly associated with longer time spent on social media in Year 4 ($\chi^2=6.72$, $p=0.021$). Lower brain activation (Year 2) in the insular cortex ($\chi^2=5.25$, $p=0.039$) and middle frontal gyrus ($\chi^2=5.09$, $p=0.042$) was associated with more time spent on social media (Year 4). However, the relationship showed a less consistent trend, with fluctuations across the activation range (Figure 4). Year 2 social media use was positively related to Year 4 social media ($\chi^2=102.55$, $p<0.01$). Sex was a significant predictor in the model, with female adolescents reporting spending more time on social media in Year 4 compared to males ($\chi^2=29.92$, $p<0.01$).

3.3 | Sleep Duration and Social Media Predicting Executive Control and Reward-Related Brain Activation

Longer sleep duration in Year 2 was associated with higher activation in the caudate in Year 4 ($\chi^2=8.18$, $p=0.033$) (Figure 5). Additionally, higher activation in Year 2 was significantly associated with higher activation in Year 4 across all regions, except for the amygdala, where the association was not significant. Social media time at Year 2 was not a significant predictor of brain activation patterns in Year 4. There were also no significant interaction effects. Age was positively associated with cingulate activation ($\chi^2=8.93$, $p=0.027$) and putamen activation ($\chi^2=8.76$, $p=0.028$), suggesting that older adolescents exhibited higher activation in those areas (Figure S1). See Tables S1–S3 for all significant and non-significant results.

3.4 | Comparison of Predictive Performance Across Analytic Approaches

Cross-validated model performance differed systematically across analytic approaches (Friedman $\chi^2(2)=24.0$, $p<0.01$). Mean cross-validated R^2 values indicated that the model predicting sleep duration in Year 4 explained the greatest proportion of variance across regions of interest (mean $R^2\approx0.14$), followed closely by the model predicting social media use in Year 4 based on categorical daily use levels (mean $R^2\approx0.13$). In contrast, the model predicting brain activation in Year 4 explained substantially less variance overall (mean $R^2\approx0.02$). Post hoc Wilcoxon signed-rank tests indicated that the sleep prediction model explained significantly more variance than the social media prediction model ($p<0.01$), although the absolute difference in explained variance was modest. Both the sleep and social media models outperformed the brain activation model (both $p<0.01$). Ranking patterns were highly consistent across regions of interest, with the sleep model most frequently achieving the highest rank, the social media model intermediate rank and the brain activation model the lowest rank.

4 | Discussion

Our findings indicate that the models predicting self-reported sleep duration explained the highest proportion of variance

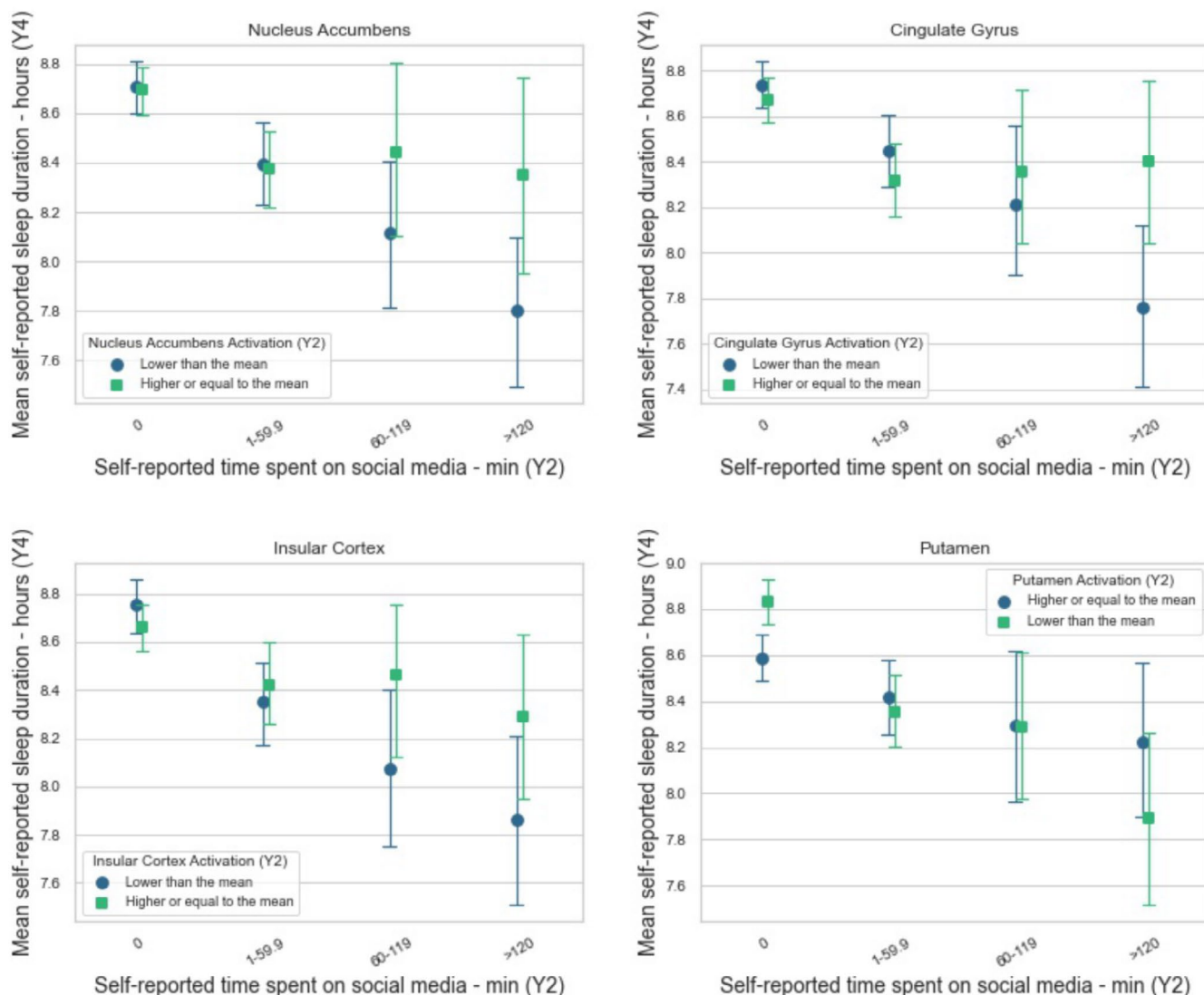


FIGURE 3 | Interaction between social media use (Y2) and brain activation (Y2) in the nucleus accumbens, cingulate gyrus, insular cortex and putamen on predicting sleep duration (Y4). Error bars represent standard errors. The time spent on social media and brain activation was categorized for visualization purposes, both variables were included as continuous measures in all models. Participants were grouped based on whether their brain activation was higher or equal to the mean (green) or lower than the mean (blue). Significant interaction effects were observed between social media use (Year 2) and activation in Year 2 within the nucleus accumbens ($\chi^2 = 7.11$, $p = 0.021$), cingulate gyrus ($\chi^2 = 6.61$, $p = 0.027$), insula ($\chi^2 = 9.15$, $p = 0.011$) and putamen ($\chi^2 = 7.13$, $p = 0.021$) on sleep duration at Year 4.

($R^2 = 0.46$), followed by those predicting self-reported social media use ($R^2 = 0.36$), with models predicting executive control and reward-related brain activation explaining the least variance ($R^2 = 0.07$). These findings suggest that sleep duration is a relatively stable behavior, influenced by prior sleep patterns, social media engagement and individual sensitivities detectable at the neural level. Adolescents with neural vulnerabilities may experience a disproportionately greater impact of social media on their sleep patterns compared to the reverse. In contrast, brain activation appears to be shaped by a broader range of factors, including maturational processes, environmental influences and measurement variability.

We found that self-reported longer sleep duration in Year 2 was associated with higher caudate activation in Year 4, suggesting a potential role of sleep in maintaining or enhancing striatal function¹⁵. This result aligns with evidence that sleep

supports brain maturation, particularly in regions implicated in executive control and reward processing (Mullin et al. 2013). However, the role of sleep in reward processing may change across adolescence, reflecting developmental shifts in neural circuitry and sleep-wake patterns. In early adolescence, Hasler et al. (2012) and Holm et al. (2009) found that insufficient sleep, later sleep onset and lower sleep quality were consistently linked to reduced caudate and striatal activation during both reward anticipation and reward reception, indicating dampened reward sensitivity. Hasler et al. (2012) reported that larger weekend-weekday advances in midsleep were associated with decreased activation in both the medial prefrontal cortex and striatum, highlighting the potential influence of circadian misalignment. By mid-adolescence, Telzer et al. (2013) observed no direct association between sleep quality and ventral striatum activation but found altered functional connectivity between reward and executive control regions, suggesting that the effects of sleep on

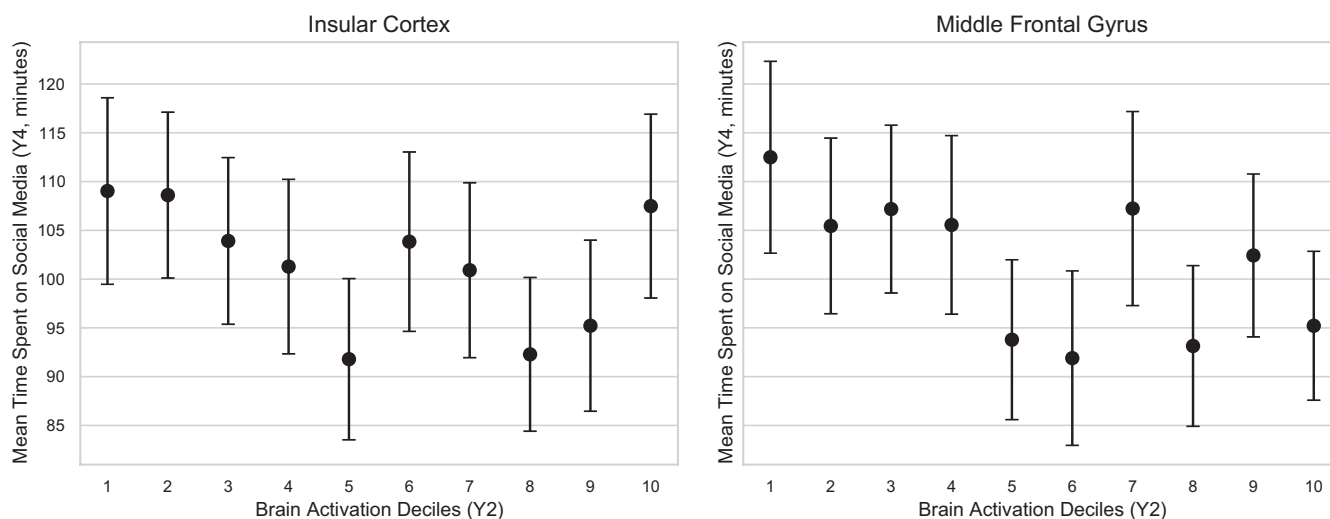


FIGURE 4 | Relationship between brain activation and social media use across deciles. The figure presents the association between brain activation in Year 2 (X-axis) and mean self-reported time spent on social media in Year 4 (Y-axis). Brain activation levels are divided into deciles (10 equal-size groups ranked from lowest to highest values) for the Insular Cortex (left panel) and Middle Frontal Gyrus (right panel). Error bars represent the standard error of the mean. Lower brain activation (Year 2) in the insular cortex ($\chi^2 = 5.25$, $p = 0.039$) and middle frontal gyrus ($\chi^2 = 5.09$, $p = 0.042$) was associated with more time spent on social media (Year 4).

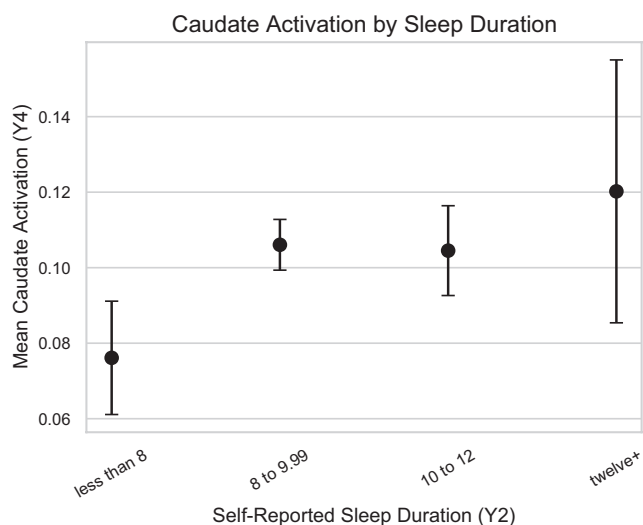


FIGURE 5 | Association between self-reported sleep duration in Year 2 and mean caudate activation in Year 4. Sleep duration was categorized for visualization purposes. The x-axis represents sleep duration categories, while the y-axis shows the mean caudate activation during the anticipation of a large reward. Error bars indicate the standard error of the mean (SEM). Longer sleep duration in Year 2 was associated with higher activation in the caudate in Year 4 ($\chi^2 = 8.18$, $p = 0.033$).

reward processing may shift as adolescents mature. In late adolescence and young adulthood, Mullin et al. (2013) found that acute sleep deprivation led to increased striatal activation, a pattern that contrasts with findings in younger adolescents and may reflect developmental changes or differences in responses to acute versus chronic sleep disturbances. These findings suggest that while sleep continues to influence reward processing across adolescence, its effects may evolve dynamically with age.

The bidirectional relationships between self-reported sleep duration and self-reported social media use align with prior

research demonstrating that insufficient sleep contributes to increased digital engagement (Semenza et al. 2022), while greater social media may disrupt sleep patterns (Hale and Guan 2015; Cain and Gradisar 2010). Additionally, the greater social media use among females (Kiss et al. 2023) is consistent with broader patterns of heightened social orientation and relational sensitivity in adolescent girls. Our findings extend this literature by showing that the relationship between social media use and subsequent sleep duration is not uniform (Perez et al. 2021), it is modulated by brain activation in key reward-related and executive control regions. Specifically, higher self-reported social media use was associated with self-reported shorter sleep duration, particularly among adolescents with lower activation in the nucleus accumbens, cingulate gyrus, insular cortex and putamen. These findings suggest that youth with lower neural engagement in these areas may be more vulnerable to the sleep-disrupting effects of social media (Holm et al. 2009), potentially due to weaker self-regulation or an increased drive for rewarding experiences (Holm et al. 2009), which contributes to their susceptibility to online reinforcement mechanisms. These results align with the ‘differential susceptibility’ framework (Belsky 2013), which suggests that individuals differ in how strongly they are affected by environmental influences—in this case, the digital environment. If adolescents with lower activation in reward-related and executive regulatory regions experience poorer sleep due to greater susceptibility to social media’s reinforcing effects, it raises the possibility that those with higher activation might engage with social media differently (Modecki et al. 2022). One plausible explanation is that adolescents with higher activation in these regions may use social media as a coping mechanism, leveraging digital engagement to regulate mood, maintain social connectedness (Modecki et al. 2022), or achieve a similar level of neural reward activation.

However, whether social media serves as an adaptive or maladaptive coping strategy in these individuals would likely depend on additional factors, such as the content they engage with

(Dziura et al. 2023), their self-regulation skills, and the overall balance between online activities and other behaviors supporting well-being. It is also possible that adolescents with higher activation do not experience the same drive for constant external stimulation and can disengage more readily when needed, reducing the potential for excessive nighttime use and its impact on sleep (Holm et al. 2009). Further research with greater granularity, ideally at a day-to-day level, is needed to better capture these individual differences and clarify how neural vulnerabilities interact with social media use and sleep patterns over time. The ABCD Study EARS data and Fitbit sub study, which includes objective measures of screen use alongside Fitbit-derived sleep patterns in future releases, may offer valuable insights into these dynamics in future studies.

The associations between lower activation in the insular cortex and the middle frontal gyrus in Year 2 with increased self-reported social media use in Year 4 highlight potential vulnerabilities in brain regions involved in interoceptive awareness, emotional regulation and executive control (Wang et al. 2008). Turel et al. (2018) reported that the posterior insula's gray matter volume was negatively correlated with social media usage symptoms, mediated by delay discounting. This finding suggests that structural and functional differences in these regions may contribute to impulsive decision-making and difficulties in regulating digital engagement over time (Crone and Konijn 2018). Given the insular cortex's role in integrating internal bodily signals (Zucker et al. 2018), lower activation may reflect reduced sensitivity to cues that typically regulate behavior, such as fatigue. Similarly, the middle frontal gyrus supports impulse control (Ding et al. 2014), and reduced engagement in this area could indicate an altered ability to disengage from online interactions (Ding et al. 2014), even when faced with negative consequences. These results align with prior research showing that heightened social feedback responsivity in the ventromedial prefrontal cortex, right inferior frontal gyrus, medial prefrontal cortex and posterior cingulate cortex may contribute to increased social media engagement (Flannery et al. 2024). Adolescents with lower activation in executive regulatory regions, such as the insular cortex and middle frontal gyrus, may be more dependent on external reinforcement (Zucker et al. 2018), relying on digital interactions to compensate for weaker endogenous reward processing (Marciano et al. 2021). Marciano et al. (2021) found that prolonged screen use was associated with increased activation in reward-related regions and decreased connectivity in cognitive control areas. Similarly, Weinstein and Lejoyeux (2020) reported heightened activation in reward regions alongside reduced activity in impulse control areas in individuals with internet gaming disorder. Over time, this pattern may reinforce excessive engagement, further altering neural regulatory mechanisms and making it harder to moderate online behaviors (Weinstein and Lejoyeux 2020). Future research with longitudinal neuroimaging and more granular assessments of social media use is needed to clarify how these neural vulnerabilities shape social media engagement across development.

5 | Limitations

First, we only used half of the Year 4 data, as more data were not yet available at the time of analysis. While we have a relatively

large sample and controlled several relevant covariates, the data may still not fully represent the broader adolescent population. Future studies with more diverse samples could help strengthen the generalizability of these findings. A second limitation is related to the self-reported nature of our sleep and social media use measures, which may be subject to biases such as social desirability and recall errors (Fadnes et al. 2009). While self-report provides valuable insight into subjective experiences, it does not always align with objective measures and does not provide precise timestamps, limiting our ability to assess the timing and patterns of sleep and social media use.

Additionally, both digital engagement and sleep health are complex behaviors characterized by multiple factors. Future studies would benefit from incorporating more detailed measures that capture not only the duration of social media use but also the type of media, content and intensity of engagement. Similarly, assessing multiple dimensions of sleep, including sleep quality, variability, timing and other physiological markers, could provide a more comprehensive understanding of how these behaviors interact over time. Although we conducted prospective analyses, it is important to note that the ABCD Study is observational, limiting our ability to draw causal inferences. Experimental or quasi-experimental approaches could help clarify the causal nature of these relationships in future research.

6 | Conclusions

Individual differences in brain activation, particularly in reward-related and executive regulatory regions, may shape susceptibility to the sleep-disrupting effects of social media. Greater social media use was linked to shorter sleep duration, and this relationship was partially explained by reduced activation in regions such as the insular cortex, middle frontal gyrus, nucleus accumbens, cingulate gyrus and putamen, suggesting that neural vulnerabilities may amplify the relationship between digital engagement on sleep. Our results also suggest the directionality between social media and sleep is stronger than vice versa, particularly among those with neural susceptibility. While some adolescents may struggle to regulate their engagement, others with higher activation in reward-related regions may use social media adaptively without experiencing the same degree of sleep disruption. Building on these novel findings may help develop targeted interventions that account for both neurobiological and behavioral factors in promoting adolescent sleep health.

Author Contributions

Orsolya Kiss: conceptualization, methodology, writing – original draft, writing – review and editing. **Linhao Zhang:** conceptualization, methodology, writing – original draft, writing – review and editing. **Bader Chaarani:** methodology, writing – review and editing. **Eva M. Müller-Oehring:** writing – original draft, writing – review and editing. **Fiona C. Baker:** writing – original draft, writing – review and editing, funding acquisition.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in ABCD data repository at <https://nda.nih.gov/study.html?id=2313>, reference number 10.15154/z563-zd24.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Mean brain activation in the cingulate gyrus and putamen (Year 4) by age (Year 2). Age was binned into 0.5-year increments for visualization. Error bars represent the standard error of the mean. **Table S1:** Predicting self-reported sleep duration in Year 4 using predictors from Year 2. **Table S2:** Predicting self-reported social media in Year 4 using predictors from Year 2. **Table S3:** Predicting brain activation in Year 4 using predictors from Year 2. **Data S1:** CFA results.